Method	Gemma-2-9B Acc	ECE	Brier	NLL	AUROC	Gemma-3-27B Acc	ECE	Brier	NLL	AUROC
Forced choice (M8)	**0.590**	0.090	**0.115**	**0.360**	**0.922**	**0.550**	0.131	**0.095**	**0.304**	**0.959**
Log-prob (no offset)	0.217	**0.037**	0.162	0.501	0.666	0.433	0.151	0.161	0.487	0.905
Log-prob (with offset)	0.217	**0.037**	0.162	0.501	0.666	0.433	0.151	0.161	0.487	0.905
Bootstrap T=0.3	0.218	0.613	0.559	6.403	0.617	0.434	0.453	0.412	5.459	0.743
Bootstrap T=0.5	0.218	0.480	0.407	2.133	0.674	0.435	0.362	0.308	2.140	0.840
Bootstrap T=0.7	0.222	0.338	0.283	0.944	0.689	0.435	0.260	0.218	0.821	0.880
Bootstrap T=1.0	0.219	0.177	0.183	0.552	0.724	0.429	0.134	0.153	0.473	0.887
Bootstrap T=1.3	0.219	0.079	0.156	0.486	0.722	0.422	0.117	0.156	0.485	0.876
Bootstrap T=1.5	0.214	0.054	0.145	0.459	0.748	0.416	0.131	0.173	0.522	0.864
Bootstrap T=1.75	0.202	0.040	0.142	0.450	0.743	–	–	–	–	–
Bootstrap T=2.0	0.204	0.050	0.145	0.455	0.738	–	–	–	–	–
Constrained label (expected)	0.217	0.463	0.374	0.958	0.671	0.433	0.318	0.312	0.833	0.821
Constrained label $P(\text{high}+)$	0.217	0.391	0.311	0.827	0.680	0.433	0.416	0.373	1.121	0.815
Constrained label $P(\text{very high})$	0.217	0.181	0.198	0.584	0.635	0.433	**0.088**	0.183	0.548	0.813
MCMC agreement T=0.125	0.217	0.722	0.700	13.66	0.546	0.436	0.527	0.510	10.27	0.592
MCMC agreement T=0.25	0.217	0.657	0.616	9.985	0.581	0.437	0.479	0.447	7.466	0.679
MCMC agreement T=0.5	0.214	0.504	0.435	3.767	0.652	0.435	0.374	0.324	3.049	0.808
Steering sensitivity	0.208	0.649	0.618	10.86	0.495	0.433	0.469	0.433	7.421	0.682
MCMC accept T=0.125	0.217	0.714	0.689	13.27	0.553	0.436	0.516	0.516	7.712	0.487
MCMC accept T=0.25	0.209	0.743	0.722	14.30	0.542	0.437	0.520	0.517	7.784	0.518
MCMC accept T=0.5	0.194	0.770	0.753	15.19	0.533	0.422	0.532	0.527	7.159	0.526
Direct (numeric)	0.217	0.783	0.783	18.02	0.500	0.433	0.562	0.559	12.10	0.531

Table 4: Full method scorecard on Gemma-2-9B and Gemma-3-27B, n=6,000 samples per row. ECE, Brier, and NLL: lower is better. Accuracy and AUROC: higher is better. Best per column in **bold**. '–': configuration not run on this oracle (the extended temperatures were run only where the ECE optimum was not interior to the original grid).

comes out weak says something about the shape of a steered oracle's answer distribution.

Background: power sampling. Karan and Du (2026) sample from the power distribution $p(x)^\alpha$ with $\alpha > 1$. The power distribution sharpens the base distribution wherever the base distribution is already peaked, and it keeps the separate modes that plain low-temperature sampling collapses into one. Karan and Du (2026) implement this as block-wise Metropolis–Hastings: the sampler proposes a continuation from a low-temperature proposal distribution, then accepts or rejects that continuation by the ratio of power-distribution densities. Two quantities of that procedure can be read as a confidence, and M6 and M7 read one each.

(M6) Power-sampling acceptance rate. We run a single block-wise Metropolis–Hastings chain on the steered oracle: B=4 blocks of 5 tokens, S=5 steps per block, and three power values $\alpha = 1/T$ with $T \in \{0.5, 0.25, 0.125\}$. Each step resamples a suffix from the low-temperature proposal and accepts it under the standard Metropolis–Hastings ratio. The confidence is the empirical acceptance rate $c_{\text{mh}} = \#\{\text{accepted}\}/(B \cdot S)$. We treated this as a speculative proxy from the start, because a high acceptance rate can also mean that the proposal distribution and the target distribution are close to each other.

The acceptance rate separates correct from wrong predictions weakly on every oracle (AUROC 0.49 to 0.60, ECE 0.52 to 0.77). Table 14 shows the mechanism. The steered oracle places almost all of its probability on one answer, the power transform concentrates that probability further, and the chain then accepts nearly every proposal whether or not the answer is correct. Lowering the proposal temperature raises task accuracy and lowers AUROC at the same time, which is what a saturating acceptance rate predicts.

(M7) Power-sampling agreement. We run k=10 independent chains with different seeds and apply the M2 mode-frequency readout to their outputs, on the same T grid as M6. Power sampling keeps the mode diversity that temperature scaling collapses, so agreement across chains could in principle sharpen the uncertain items.

At $\alpha = 2$ this reaches AUROC 0.858 on Qwen3.6-27B and 0.808 on Gemma-3-27B, which is close to the bootstrap. Its ECE stays well above the bootstrap's on every oracle, and it costs roughly five times the bootstrap's wall-clock time (table 15). The α sweep in table 16 reverses the M6 pattern: at $\alpha = 2$ the chains stay diverse, and at $\alpha = 8$ all ten chains collapse onto the greedy trajectory, so the mode frequency saturates at 1.0 whether or not the answer is correct. The best α for agreement is therefore small, which is the same direction as the bootstrap optimum. Power sampling was built for

Method	ECE	Brier	NLL	AUROC
Qwen3-8B				
Forced choice (M8)	.078 [.070,.086]	.100 [.096,.104]	.324 [.314,.333]	.940 [.934,.946]
Log-prob (with offset)	.255 [.244,.266]	.246 [.238,.254]	.730 [.707,.754]	.840 [.831,.850]
Bootstrap T=1.0	.057 [.048,.067]	.163 [.159,.168]	.498 [.488,.509]	.829 [.819,.839]
Bootstrap T=1.5	.083 [.076,.093]	.173 [.168,.178]	.522 [.510,.533]	.824 [.813,.836]
Constrained label $P(\text{very high})$	.425 [.413,.437]	.387 [.378,.395]	1.064 [1.040,1.087]	.812 [.802,.823]
Direct (numeric)	.582 [.571,.595]	.580 [.569,.593]	12.658 [12.397,12.938]	.516 [.510,.521]
Qwen3.6-27B				
Forced choice (M8)	.080 [.073,.088]	.100 [.096,.104]	.332 [.323,.342]	.936 [.930,.943]
Log-prob (with offset)	.131 [.122,.141]	.158 [.151,.166]	.489 [.469,.511]	.835 [.822,.847]
Bootstrap T=1.0	.147 [.139,.156]	.130 [.127,.134]	.431 [.423,.439]	.851 [.839,.864]
Bootstrap T=1.5	.103 [.095,.112]	.129 [.125,.133]	.422 [.412,.432]	.812 [.798,.826]
Constrained label $P(\text{very high})$	.069 [.061,.079]	.158 [.153,.165]	.478 [.464,.493]	.778 [.764,.790]
Direct (numeric)	.753 [.741,.764]	.752 [.742,.763]	16.022 [15.756,16.293]	.404 [.391,.416]
Gemma-2-9B				
Forced choice (M8)	.090 [.082,.098]	.115 [.111,.118]	.360 [.352,.369]	.922 [.915,.929]
Log-prob (with offset)	.037 [.031,.047]	.162 [.156,.168]	.501 [.485,.517]	.666 [.650,.682]
Bootstrap T=1.0	.177 [.167,.186]	.183 [.179,.187]	.552 [.541,.564]	.724 [.708,.739]
Bootstrap T=1.5	.054 [.048,.064]	.145 [.141,.150]	.459 [.449,.470]	.748 [.733,.761]
Constrained label $P(\text{very high})$	.181 [.173,.193]	.198 [.194,.202]	.584 [.575,.593]	.635 [.616,.653]
Direct (numeric)	.783 [.772,.794]	.783 [.772,.794]	18.022 [17.780,18.275]	.500 [.500,.500]
Gemma-3-27B				
Forced choice (M8)	.131 [.124,.138]	.095 [.092,.099]	.304 [.295,.313]	.959 [.954,.964]
Log-prob (with offset)	.151 [.142,.160]	.161 [.157,.165]	.487 [.477,.497]	.905 [.897,.912]
Bootstrap T=1.0	.134 [.124,.144]	.153 [.150,.157]	.473 [.464,.482]	.887 [.879,.896]
Bootstrap T=1.5	.131 [.121,.141]	.173 [.169,.177]	.522 [.512,.531]	.864 [.854,.873]
Constrained label $P(\text{very high})$	.088 [.078,.099]	.183 [.180,.187]	.548 [.541,.556]	.813 [.802,.823]
Direct (numeric)	.562 [.548,.575]	.559 [.545,.572]	12.102 [11.794,12.394]	.531 [.526,.536]

Table 5: Bootstrap 95% CIs (1000 resamples) for the main scorecard rows on all four oracles. Point estimate followed by the 2.5/97.5 percentiles. n=6,000 per row.

problems where several distinct answers genuinely compete, as in long reasoning chains. Our task has one correct answer and a sharply peaked oracle, which is the regime where power sampling has the least to add.

(M8) Steering-coefficient sensitivity. Zur et al. (2025) report that token-level uncertainty correlates with how easily a steering intervention flips a model's output, so a confident model is hard to push and an unsure one is easy. The oracle version varies the strength of the injection: we greedy decode at steering coefficients $s \in \{0.5, 0.75, 1.0, 1.25, 1.5\}$ around the default s=1.0 and report the modal answer with confidence equal to its mode frequency. An activation the oracle reads clearly should decode to the same word at every nearby s.

The reading is coarse by construction. With five injection strengths the mode frequency can take only the five values $\{0.2, 0.4, 0.6, 0.8, 1.0\}$, which is too few levels to express a probability (ECE 0.40 to 0.65). The ranking signal is present on three of the four oracles (AUROC 0.763 on Qwen3-8B, 0.725 on Qwen3.6-27B, 0.682 on Gemma-3-27B) and sits at chance on Gemma-2-9B (0.495). The resolution is what limits the calibration, so the method stays usable as a yes-or-no stability check.

Table 15 compares power-sampling agreement at its best temperature against bootstrap at its best original-grid temperature on each Qwen oracle. Bootstrap has lower ECE and Brier on both; agreement is $+0.007$ higher in AUROC on Qwen3.6-27B and -0.026 lower on Qwen3-8B, at roughly $5\times$ the wall-clock cost.

K Constrained-label elicitation details

The constrained-label method (M4) reuses the answer turn of M3 and replaces the free-form confidence turn with "Reply with exactly one of: very low, low, medium, high, very high". Each of the five labels is teacher-forced after the prompt and scored by its joint token log-probability under the steering hook; the five scores are softmax-normalized into a label distribution. Readouts: *expected* maps the labels to $\{0.1, 0.3, 0.5, 0.7, 0.9\}$ and takes the probability-weighted mean; $P(\text{very high})$ is the mass on the top label; $P(\text{high+})$ is the mass on the top two. The modal label is "very high" on nearly every sample on every oracle; the ranking signal is carried by the distribution, and the modal label alone

Method	Word-disjoint split					Random 50/50 split				
	Uncal	Temp	Platt	Iso	Beta	Uncal	Temp	Platt	Iso	Beta
Forced choice (M8)	.076	.041	.040	.043	.041	.081	.037	.016	.012	.014
Log-prob (with offset)	.172	.180	.128	.127	.128	.258	.157	.018	.021	.017
Bootstrap T=0.7	.156	.159	.118	.113	.111	.092	.089	.029	.025	.038
Bootstrap T=1.0	.061	.033	.094	.094	.095	.067	.031	.024	.029	.024
Bootstrap T=1.5	.069	.072	.078	.077	.077	.087	.093	.020	.019	.015
Constrained label $P(\text{very high})$	.525	.295	.152	.149	.151	.423	.107	.026	.016	.016
Direct (numeric)	.691	.354	.219	.217	.217	.579	.241	.009	.007	.007
MCMC accept T=0.125	.642	.327	.197	.197	.197	.541	.223	.007	.012	.010
MCMC agreement T=0.5	.334	.274	.145	.135	.133	.261	.176	.060	.023	.024
Steering sensitivity	.485	.307	.136	.135	.135	.399	.172	.013	.013	.016

Table 6: Post-hoc calibration on Qwen3-8B: test-set ECE after fitting each calibrator on the fit slice. *Word-disjoint*: fit on 10 of 20 secret words, evaluate on the other 10. *Random 50/50*: random sample-level split, identical across methods. Lower is better.

Method	Word-disjoint split					Random 50/50 split				
	Uncal	Temp	Platt	Iso	Beta	Uncal	Temp	Platt	Iso	Beta
Forced choice (M8)	.088	.055	.036	.029	.025	.090	.054	.024	.027	.022
Log-prob (with offset)	.142	.090	.030	.017	.021	.138	.096	.030	.019	.021
Bootstrap T=0.7	.209	.156	.026	.016	.030	.210	.160	.024	.017	.034
Bootstrap T=1.0	.145	.017	.012	.019	.012	.152	.024	.027	.022	.026
Bootstrap T=1.5	.101	.054	.016	.022	.020	.103	.042	.014	.010	.007
Constrained label $P(\text{very high})$	.084	.077	.057	.030	.027	.073	.070	.041	.017	.031
Direct (numeric)	.740	.399	.035	.028	.042	.744	.403	.017	.017	.016
MCMC accept T=0.125	.728	.404	.023	.023	.023	.733	.409	.014	.014	.015
MCMC agreement T=0.5	.381	.319	.044	.022	.022	.380	.319	.041	.016	.016
Steering sensitivity	.491	.306	.027	.025	.025	.494	.315	.017	.018	.018

Table 7: Post-hoc calibration on Qwen3.6-27B: test-set ECE after fitting each calibrator on the fit slice. *Word-disjoint*: fit on 10 of 20 secret words, evaluate on the other 10. *Random 50/50*: random sample-level split, identical across methods. Lower is better.

would reproduce the free-form self-report. In the n=30 pilot of the earlier version, $P(\text{True})$ scoring (Kadavath et al., 2022) was also tested and did not separate the pools (AUROC 0.440), because the model assigns high probability to "yes" whenever the proposed answer is a plausible secret-word phrase regardless of activation match.

L Training the secret-word target models

The Qwen3-8B target adapters are the released ones of Karvonen et al. (2025) and the Gemma-2-9B target adapters are those of Cywiński et al. (2025). We train the Qwen3.6-27B and Gemma-3-27B adapters ourselves, one per secret word, with the recipe below.

The training data for one word is the corresponding per-word conversation set of Cywiński et al. (2025), in which the assistant hints at the word and declines direct requests to name it. We expand every conversation into one row per assistant turn, each row ending at the turn it supervises. The expansion exists because the Qwen chat templates emit a thinking block only on the final assistant turn, so supervising earlier turns in place would train on a rendering the model never sees at inference. We hold out 10% of the rows for evaluation, then mix the training rows one to one with first turns drawn from UltraChat and filtered to the secret-word set's maximum character length. Without that mixture the target stops behaving as a general assistant, which would make the collected activations unrepresentative.

Each adapter is LoRA with rank 32, α=64, and dropout 0.05, attached to every linear layer of the text stack. Gemma-3 and Qwen3.6 need an explicit target pattern here for the same reason described in section M: an "all linear" selection would also wrap the vision tower and, on Qwen3.6, the Gated DeltaNet routing projections, none of which receive gradients during text-only training. We train with learning rate 5×10^{-5} on a constant schedule with 5% warmup, weight decay 0.01, bfloat16, and a maximum sequence length of 1024, for up to 10 epochs with early stopping after two evaluations without improvement in validation loss, restoring the best checkpoint. One adapter takes roughly two hours on a single B200.

Method	Word-disjoint split					Random 50/50 split				
	Uncal	Temp	Platt	Iso	Beta	Uncal	Temp	Platt	Iso	Beta
Forced choice (M8)	.122	.072	.097	.093	.092	.092	.031	.019	.015	.014
Log-prob (with offset)	.072	.094	.101	.100	.102	.046	.040	.026	.021	.027
Bootstrap T=0.7	.391	.362	.121	.110	.115	.344	.294	.033	.027	.032
Bootstrap T=1.0	.225	.199	.099	.099	.099	.178	.148	.010	.021	.021
Bootstrap T=1.5	.096	.074	.095	.096	.095	.056	.018	.014	.013	.014
Constrained label $P(\text{very high})$	.243	.225	.128	.121	.118	.185	.172	.030	.011	.017
Direct (numeric)	.852	.518	.139	.139	.139	.789	.455	.013	.013	.013
MCMC accept T=0.125	.780	.477	.138	.137	.137	.722	.416	.013	.013	.013
MCMC agreement T=0.5	.564	.396	.130	.128	.128	.513	.335	.014	.015	.015
Steering sensitivity	.716	.458	.138	.138	.137	.658	.399	.018	.017	.017

Table 8: Post-hoc calibration on Gemma-2-9B: test-set ECE after fitting each calibrator on the fit slice. *Word-disjoint*: fit on 10 of 20 secret words, evaluate on the other 10. *Random 50/50*: random sample-level split, identical across methods. Lower is better.

Method	Word-disjoint split					Random 50/50 split				
	Uncal	Temp	Platt	Iso	Beta	Uncal	Temp	Platt	Iso	Beta
Forced choice (M8)	.128	.090	.016	.014	.013	.133	.093	.012	.015	.017
Log-prob (with offset)	.184	.170	.040	.038	.038	.152	.146	.021	.024	.018
Bootstrap T=0.7	.231	.193	.098	.029	.046	.262	.222	.094	.028	.043
Bootstrap T=1.0	.123	.087	.035	.038	.035	.136	.106	.020	.024	.023
Bootstrap T=1.5	.162	.152	.047	.047	.047	.128	.121	.015	.016	.014
Constrained label $P(\text{very high})$	.081	.045	.085	.091	.087	.094	.045	.024	.021	.023
Direct (numeric)	.512	.176	.110	.109	.110	.566	.230	.009	.009	.009
MCMC accept T=0.125	.464	.113	.103	.103	.103	.519	.169	.009	.009	.009
MCMC agreement T=0.5	.338	.208	.050	.054	.055	.378	.189	.034	.009	.013
Steering sensitivity	.425	.138	.089	.090	.090	.471	.189	.003	.003	.003

Table 9: Post-hoc calibration on Gemma-3-27B: test-set ECE after fitting each calibrator on the fit slice. *Word-disjoint*: fit on 10 of 20 secret words, evaluate on the other 10. *Random 50/50*: random sample-level split, identical across methods. Lower is better.

M Training the Qwen3.6-27B activation oracle

The Qwen3-8B oracle is the released checkpoint of Karvonen et al. (2025). For Qwen3.6-27B no oracle existed prior to this work. We adapt their LatentQA trainer to a hybrid Gated DeltaNet ($3/4$ of token-mixing sublayers) plus Gated Attention ($1/4$ of sublayers) architecture and train a verbalizer with the same data mixture (system-prompt question answering, binary classification, self-supervised context prediction) and the same LoRA recipe (r=8, one epoch, bf16). Two changes were needed:

LoRA target selection. The upstream regex matches only `q_proj|k_proj|v_proj|o_proj` and the MLP projections. Those names exist only in the $1/4$ full-attention sublayers of Qwen3.6. The $3/4$ Gated DeltaNet sublayers expose Linear modules under different names (`in_proj_qkv`, `in_proj_z`, `in_proj_b`, `in_proj_a`, `out_proj`). Without expanding the regex, LoRA wraps only one quarter of the token-mixing capacity and training loss flattens well above the homogeneous baseline. Our expanded regex covers both attention types while still excluding the vision tower and `lm_head`.

Activation-collection slicing. The upstream multi-layer activation collector had an off-by-one in its offset window. We patched the slicing to honor `[max_offset:min_offset]` consistently, which matters when collecting from the trailing tokens of long inputs.

After these two fixes the verbalizer trains to comparable out-of-distribution validation loss as the homogeneous Qwen3-8B oracle (one B200, $\approx$ 36 hours wall-clock, single epoch on $\approx$ 1M examples). The matching secret-word target adapters are trained with the recipe of section L.

N Hardware, software, and runtime

All runs use a single NVIDIA B200 GPU with bfloat16 weights. The Qwen3-8B benchmark sweep completes in $\approx$ 8 hours wall-clock; the Qwen3.6-27B sweep in $\approx$ 48 hours; the two Gemma sweeps and the forced-choice, constrained-label, and extended-temperature backfills add $\approx$ 70 hours combined. Software: PyTorch 2.11, Hugging Face Transformers 5.8, PEFT 0.19, flash-attn